📅 Working with Dates in SQL – Simple Notes

🔷 Why Are Dates Important?

Dates are used to:
✓ Track trends over time
✓ Generate monthly/yearly reports
✓ Analyze customer behavior
✓ Make business decisions

👉 Date handling is very important in Data Analysis.

📂 Sample Column: purchase_date

Example format:

2023-01-05

Format:
YYYY-MM-DD

🔎 1️⃣ Extracting Date Components

Sometimes we need only:

- Day
- Month
- Year

🛠 Step 1: Add Columns

```
ALTER TABLE messy_indian_dataset
  ADD COLUMN day INT,
  ADD COLUMN month INT,
  ADD COLUMN year INT;
```

👉 Adds new columns to store extracted values.

🛠 Step 2: Update Columns

```
UPDATE messy_indian_dataset
SET day = DAY(purchase_date),
  month = MONTH(purchase_date),
```

```
year = YEAR(purchase_date);
```

🔷 **Functions Used**

Function	Purpose
DAY()	Extracts day
MONTH()	Extracts month number
YEAR()	Extracts year

📌 **Example:**

If purchase_date = 2023-01-05

Then:

- Day → 5
- Month → 1
- Year → 2023

📅 2️⃣ **Adding Day of the Week**

Sometimes we need the **name of the day** (Monday, Tuesday, etc.)

🛠 **Step 1: Add Column**

```
ALTER TABLE messy_indian_dataset
ADD COLUMN day_of_week VARCHAR(10);
```

🛠 **Step 2: Update Column**

```
UPDATE messy_indian_dataset
SET day_of_week = DAYNAME(purchase_date);
```

🔷 **Function Used**

Function	Purpose
DAYNAME()	Returns weekday name

📌 **Example:**

If date = 2023-01-05
Output → Thursday

🗓️ 3️⃣ Adding Month Name

Instead of month number (1, 2, 3), we may want:
January, February, March

🛠️ Step 1: Add Column

```
ALTER TABLE messy_indian_dataset

ADD COLUMN month_name VARCHAR(10);
```

🛠️ Step 2: Update Column

```
UPDATE messy_indian_dataset

SET month_name = MONTHNAME(purchase_date);
```

🔷 Function Used

Function	Purpose
MONTHNAME()	Returns month name

📌 Example:

If date = 2023-01-05
Output → January

🎯 Why Extract Date Components?

✓ Monthly sales analysis
✓ Yearly performance reports
✓ Weekday vs Weekend sales
✓ Trend analysis
✓ Time-based grouping

🚀 Important Date Functions (Quick Revision Table)

Function	Description
NOW()	Current date & time

Function	Description
CURDATE()	Current date
DAY()	Extract day
MONTH()	Extract month
YEAR()	Extract year
DAYNAME()	Weekday name
MONTHNAME()	Month name

💡 **Best Practice**

Instead of permanently adding columns, you can also extract date parts directly in SELECT:

```
SELECT
  purchase_date,
  DAY(purchase_date) AS day,
  MONTH(purchase_date) AS month,
  YEAR(purchase_date) AS year
FROM messy_indian_dataset;
```

👉 Cleaner and more efficient in real-world projects.

🎯 **Final Key Concept**

Working with dates in SQL helps you:

1️⃣ Break down time data
2️⃣ Perform time-based analysis
3️⃣ Generate better reports

Time-based analysis = Powerful insights 📊